

Counting

a. Permutations and Combinations

Permutations

- Def. A **permutation** of a set is an **ordered** arrangement of its elements (a list with the same elements as the set)
- Ex. Suppose we consider the set $\{1,2,3\}$ -- this has 6 permutations
 - $\{1,2,3\}, \{1,3,2\}, \{2,1,3\}, \{2,3,1\}, \{3,1,2\}, \{3,2,1\}$
- How many permutations of a set of n elements? $n!$
- Can also consider **2-permutations** (i.e., ordered subsets of size two) of the same set – this also has size 6
 - $\{1,2\}, \{1,3\}, \{2,1\}, \{2,3\}, \{3,1\}$ and $\{3,2\}$
- And more generally k – *permutations*, for $k \leq n$
- The fact that we got the same number in both examples is not a coincidence!

Number of r – Permutations

- Def. Let $P(n,r)$ denote the number of r -permutations of an n element set
- Thm. $P(n,r) = n \cdot (n - 1) \cdot \dots \cdot (n - (r - 1))$
- Pf. There are n ways to choose the first, $n-1$ remaining ways to choose the second and so on.
- Cor. If n is positive, and $0 \leq r \leq n$, then $P(n,r) = \frac{n!}{(n-r)!}$
- Pf 2. Start with the number of ways to order an n element set, this is $n!$.
 - But we could have counted differently: first choose r items from the set – there's $P(n,r)$ different ways to do this.
 - Then choose the remaining $n - r$ items in $(n - r)!$ Different ways.
 - So we have $n! = P(n,r) \cdot (n - r)! \Rightarrow P(n,r) = \frac{n!}{(n-r)!}$

Permutations example

- Ex: How many permutations of the letters ABCDEFGH contain the letters ABC consecutively?
 - There's a trick here! We view the underlying set as consisting of 6 elements: ABC, D, E, F, G, and H.
 - Any permutation that has the characters we want amounts to a permutation of these six
 - So the answer is $P(6,6)=6!=720$.

Combinations

- Closely related problem is how many distinct subsets of size r can be formed from an n element set.
- Each such subset is called an **r-combination**
- Def. An r -combination of a set with n elements is a subset of size r . We denote the number of such subsets as **$C(n,r)$**
- How are *combinations* different than *permutations*?
 - A permutation refers to ordered arrangements of elements (e.g., $\{1,2,3\}$ is distinct from $\{3,2,1\}$).
 - A combination is just a subset (so ordering within the subset is ignored)
- Are there more **r-permutations** of an n element set or **r-combinations**?
 - In general there are more **r-permutations** of an n element set, than **r-combinations** from that set.

Number of combinations

- **Thm.** If n is positive, and $0 \leq r \leq n$, then $C(n, r) = \frac{n!}{r!(n-r)!}$
- **Intuition:** Recall there are $n!/(n-r)!$ r -permutations of a set of size n .
 - But when comparing with r -combinations, we've *overcounted* since many r -permutations correspond to the same r -combination.
 - Easy to see we've overcounted by a factor of $r!$, why?
 - Because there are $r!$ r -permutations per each r -combination
- **Pf.** Let's just formalize this intuition:
 - $P(n, r) = C(n, r) \cdot P(r, r)$
 - $\Rightarrow C(n, r) = \frac{P(n, r)}{P(r, r)} \Rightarrow C(n, r) = \frac{n!}{r!(n-r)!}$

Combinations example

- It's traditional to consider examples with decks of cards
- However, if you're like me, you don't play card games very often
- So let's review the basics:
 - There's 52 total cards (not counting Jokers)
 - There are 13 "denominations" or "kinds" (i.e., 2,...,9,10,Jack, Queen, King, Ace)
 - There are four "suits" (clubs, diamonds, hearts, spades)
 - A poker hand consists of 5 cards
- Ex. How many distinct poker hands exist?
 - Answer: $C(52,5) = 2,598,960$

More card counting examples

- **Reminder:** A full house is a poker hand (5 cards) that:
 - consists of a three-of-a-kind (i.e., a triple of the same kind, possibly different suits)
 - together with a pair (two of the same kind) – but can be different kind from the above triple
- **Ex.** How many distinct “full houses” exist?
 - Pick two kinds in order (one for the triple, one for the pair)
 - There's $P(13,2)$ ways to do this – why not $C(13,2)$?
 - Now we have narrowed our options down to the same kind, but there are four cards of the same denomination in each deck (one for each suit)
 - So there's $C(4,3)$ different ways to choose the triple and $C(4,2)$ ways to choose the pair.
 - So there's $n = P(13,2) \cdot C(4,3) \cdot C(4,2)$

Even more card examples!

- Recall, a flush is a poker hand (5 cards), in which all cards are the same suit
- Ex. How many flushes exist?
 - First there are $C(4,1) = 4$ different suits
 - Then from that suit there are $C(13,5)$ different hands in that suit
 - So answer $n = C(4,1) \cdot C(13,5) = 5,148$

Combinatorial proofs

- Def. A combinatorial proof is a proof based on a counting argument.
 - (Generally these proofs work by counting the same thing two different ways to achieve combinatorial identity)
- Thm. For all $n > 0$ and $0 \leq r \leq n$, $C(n, r) = C(n, n - r)$
- Pf. Given a set of n elements there's two ways to pick r elements.
 - Can enumerate its r elements subsets directly: get $C(n, r)$ or
 - Can enumerate their complements i.e., “throw out” $n - r$ elements, $C(n, n - r)$ different ways – now we're left with r elements!